

AS • Edexcel • Physics

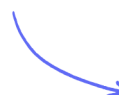
🕒 24 mins ❓ 3 questions

Structured Questions

Resistance, Resistivity & Potential Dividers

Electrical Resistivity / Core Practical 2: Investigating Resistivity / Current & Drift Velocity / Potential Difference & Conductor Length / Potential Dividers / Potential Dividers & Variable Resistance

Scan here to return to the course
or visit [savemyexams.com](https://www.savemyexams.com)



Total Marks

/24

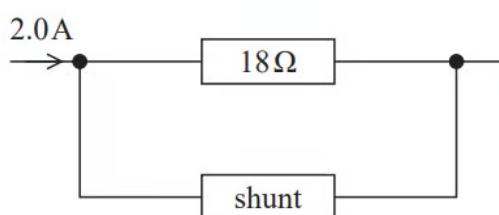
1 (a) Analogue ammeters were used before digital meters became widely available.

The analogue ammeter shown will measure a maximum current of 1.0 mA and has a resistance of $18\ \Omega$.



(Source: © David J. Green/Alamy Stock Photo)

The analogue ammeter can be adapted to measure a larger current by adding a resistor, known as a shunt, in parallel with the ammeter. The arrangement is shown below. The analogue ammeter is represented by the $18\ \Omega$ resistor.



The maximum current through the $18\ \Omega$ resistor remains as 1.0 mA.

Show that the shunt would need to have a resistance of about $0.01\ \Omega$ to adapt this ammeter to read up to a maximum current of 2.0 A.

(3 marks)

(b) A shunt of this resistance was usually made from Manganin wire.

Calculate the length of Manganin wire of radius 0.95 mm required to make this shunt.
resistivity of Manganin = $4.55 \times 10^{-7} \Omega \text{ m}$

Length =

.....

.....

.....

(3 marks)

- 2 (a)** A “metre bridge” is a circuit which can be used to measure an unknown resistance accurately. The metre bridge includes a metre length of nichrome wire.

Calculate the resistance of a 1.00 m length of the nichrome wire.

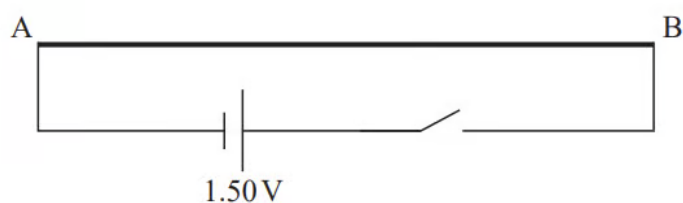
resistivity of nichrome = $1.12 \times 10^{-6} \Omega \text{ m}$

diameter of wire = $4.00 \times 10^{-4} \text{ m}$

Resistance =

(3 marks)

- (b)** This metre length of wire, labelled AB, is connected to a 1.50 V cell of negligible internal resistance and a switch as shown.



- i) Explain how the potential along this wire varies with distance from A when the switch is closed.

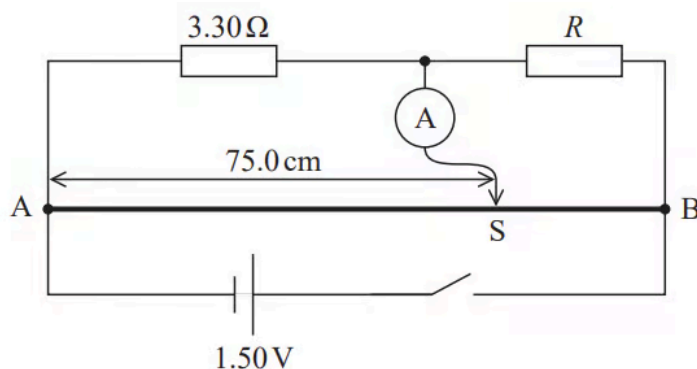
(2)

- ii) Show that the potential difference between A and a point 75.0 cm along the wire from A is about 1.1 V.

(2)

(4 marks)

- (c) The metre bridge circuit is shown. The circuit includes a resistor of resistance $3.30\ \Omega$, a very sensitive ammeter and a resistor of unknown resistance R .



A metal slider S can be moved along the nichrome wire and pressed firmly against it to make an electrical connection.

When the switch is closed and S is 75.0 cm along the nichrome wire, the ammeter reads 0 A because the potential difference across the ammeter is zero.

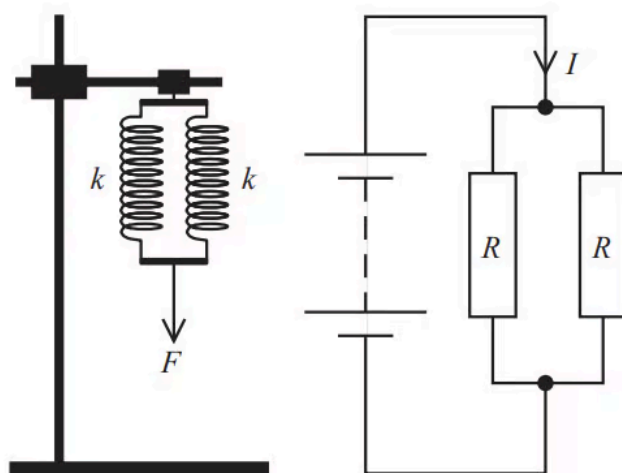
Calculate R .

$R = \dots\dots\dots$

(2 marks)

- 3 (a)** A student is experimenting with different combinations of springs and recalls that in physics it is often possible to model different physical situations in similar ways.

The student suggests that a parallel combination of springs could be a model for a parallel combination of resistors in a circuit.



Derive an expression for the effective resistance R_{eff} of two resistors R_1 and R_2 connected in parallel in a circuit.

(3 marks)

- (b)** Assess the validity of the student's suggestion by considering the effective stiffness of two identical springs in parallel.

(6 marks)